

AuroraPhoto

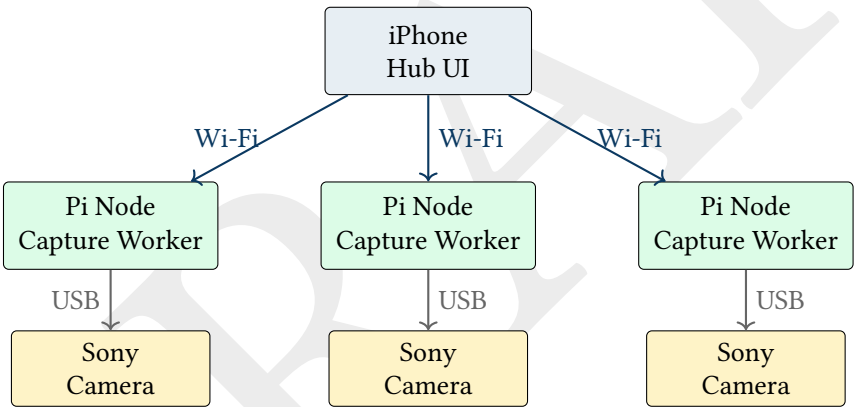
Distributed Edge Control for Adaptive Aurora Capture and Star Sharpness

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AuroraPhoto Project

March 2026 • Version 0.1.0

Last updated: March 12, 2026



Abstract. AuroraPhoto is a distributed astrophotography control system aimed at capturing high-quality aurora and night-sky imagery under rapidly changing conditions. The core design uses an iPhone as a field-ready coordination hub and one or more Raspberry Pi capture nodes connected to Sony Alpha cameras over USB. Each node performs local (edge) image analysis to compute star sharpness (Half-Flux Radius, HFR) and exposure histograms from preview imagery, enabling low-latency focus correction and adaptive exposure profile switching during aurora “bursts”. This report specifies the target architecture, key control loops, failure modes specific to cold-field USB operation, and a roadmap-to-requirements traceability model suitable for implementation and validation.

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Acronyms and Terms

Term	Meaning
HFR	Half-Flux Radius; a star sharpness proxy computed from extracted point sources. Lower is sharper.
SNR	Signal-to-noise ratio; proxy for whether stars can be reliably measured in previews.
p95/p99	95th/99th percentile of a latency distribution; used to reason about tail latency.
Delta-trigger	Trigger scheduling method where nodes receive a command to act at a future timestamp to absorb network jitter.

Table 1. Acronyms and terms used in this report.

1 Problem Framing

Aurora photography presents an unusually adversarial control environment: scene illumination changes rapidly; the most visually valuable structure (discrete rays and curtains) can require short exposures; focus drifts with temperature; and field constraints (cold, battery, unreliable links) magnify small failure modes.

Design Decision 1: Edge-first Control

The system performs all time-critical perception (histogram checks, star sharpness) on the Raspberry Pi node. The phone coordinates, visualizes, and triggers, but does not become a compute bottleneck.

Insight:
The key product value is not “automation” as in “unattended intervalometer capture”; it is closed-loop control where preview analysis feeds back into exposure and focus decisions within seconds.

2 System Model and Component Boundaries

We model AuroraPhoto as a hub-and-spoke distributed control system consisting of three primary roles: a mobile coordination hub, one or more edge capture nodes, and camera devices.

2.1 Component Responsibilities

Component	Primary Responsibility	Interface
iPhone Hub	UI, settings, composition aids (compass/overlay), coordinated triggering, status dashboard	Wi-Fi
Pi Capture Node	Camera control, preview analysis (HFR, histogram), local buffering, watchdog/recovery	USB + Wi-Fi
Sony Camera	RAW acquisition, focus motor control, exposure parameter execution	USB (PTP/SDK)

Table 2. AuroraPhoto component boundaries and responsibilities.

2.2 Interfaces and Constraints

Two constraints dominate the architecture:

1. Preview-to-decision latency must be low (seconds) to preserve aurora beam definition.
2. Bandwidth is scarce in the field; RAW transfer is prohibitively expensive for real-time loops.

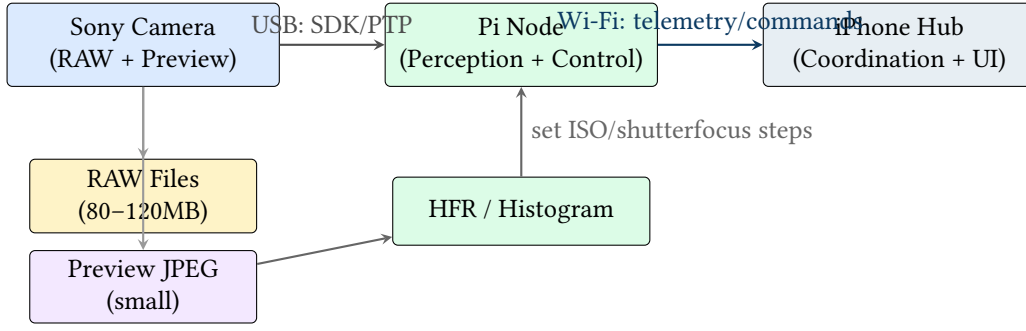


Figure 1. Data movement is asymmetric: previews drive decisions; RAW is buffered and transferred opportunistically.

3 Control Loops

AuroraPhoto is best described as three interacting control loops running at different time scales.

3.1 Coupled-loop View (Focus, Exposure, Burst)

The loops are coupled through shared observables (preview frames) and shared actuators (camera parameters). A useful mental model is a triad of controllers operating on different objectives but competing for the same camera time.

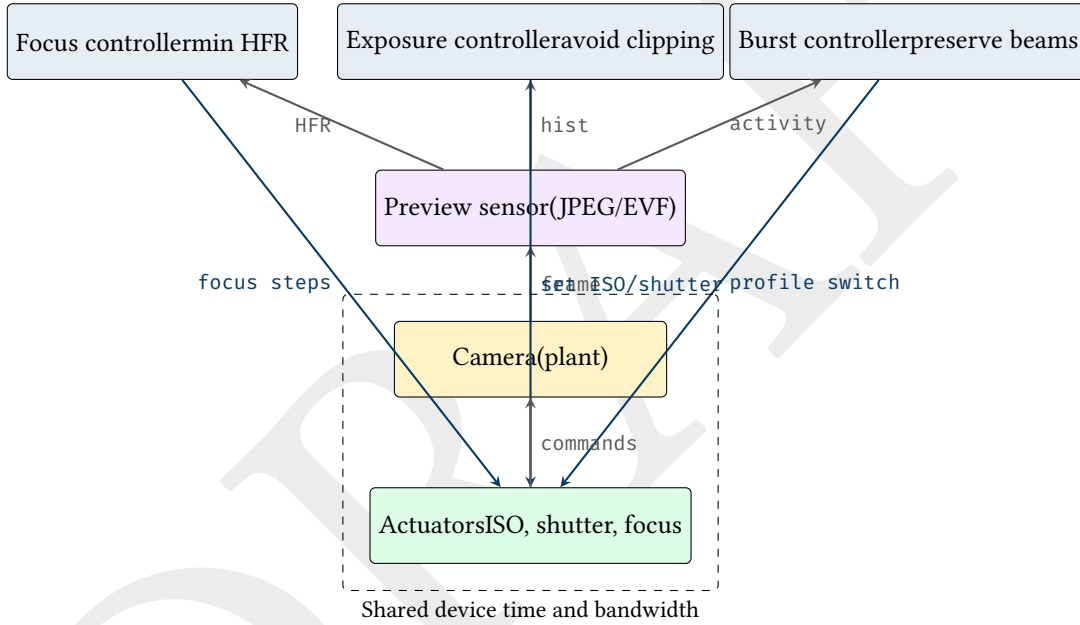


Figure 2. Coupled-loop view: three controllers observe the same preview stream and compete for shared actuation bandwidth.

3.2 Focus Loop (HFR Minimization)

The focus loop uses preview frames to detect stars and compute a scalar sharpness metric (HFR). The control objective is to minimize HFR by stepping the focus motor.

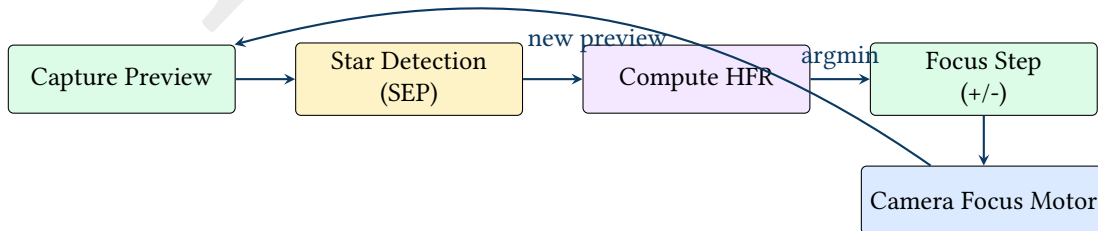


Figure 3. Focus loop: preview-driven HFR minimization on the edge node.

3.3 Exposure Loop (Histogram Guard Rails)

Aurora scenes often clip the green channel; once clipped, beam structure is irrecoverable.

: Green-channel Clip Avoidance

Treat preview histogram clipping as a hard constraint: if any channel exceeds a clipping threshold, shorten shutter and/or reduce ISO before continuing the sequence.

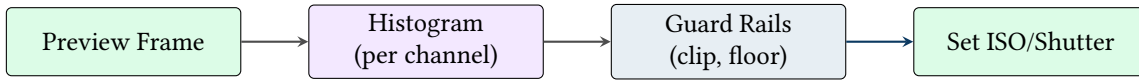


Figure 4. Exposure loop: histogram-based constraint enforcement to protect highlight detail.

3.4 Burst Loop (Activity-driven Profile Switching)

Burst logic switches between exposure profiles (e.g., “wait” vs “burst”) based on detected aurora activity. In v1, a heuristic can be histogram-based; in v2, ML classification is targeted.

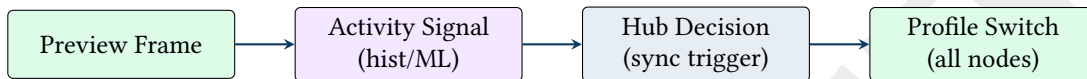


Figure 5. Burst loop: activity detection escalates capture aggressiveness while preserving synchronization.

4 Distributed Synchronization

Multi-node capture requires synchronized triggering while tolerating intermittent connectivity.

Design Decision 2: Delta-triggering

Nodes maintain a shared schedule but wait for a “delta” trigger message from the hub. This reduces sensitivity to clock skew while enabling aligned bursts.

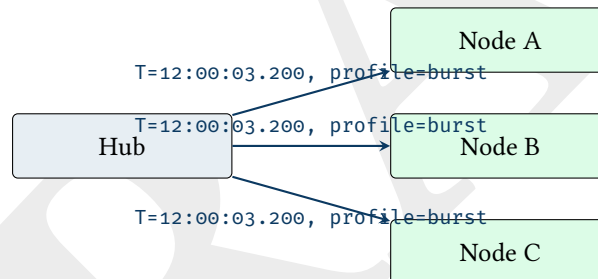


Figure 6. Delta-triggering: a hub message specifies a future trigger timestamp to absorb network jitter.

5 Reliability and Field Failure Modes

Field conditions expose failure classes not common in lab camera automation.

5.1 Sony USB “Hang” and Recovery

Sony cameras can appear present on the USB bus yet be unresponsive to PTP commands. The recovery strategy is to detect timeouts and power-cycle the relevant USB port (via a hub with per-port power switching).

USB Hang Risk

If recovery requires a physical unplug/replug, the system ceases to be “remote”. USB reset via controllable hubs is a design requirement (SHRP-04).

5.2 Thermal Focus Drift

Even if focus is correct at startup, temperature-driven drift can invalidate the imaging run. Monitoring HFR provides a quantitative trigger for refocus routines.

6 Requirements Traceability

7 End-to-End Operational Scenarios

This section describes the primary user-visible workflows as message-and-control sequences. The intent is to make the architecture testable: each scenario implies timing expectations, failure handling, and a minimum set of state transitions.

7.1 Scenario A: Field Bring-up and Node Enrollment

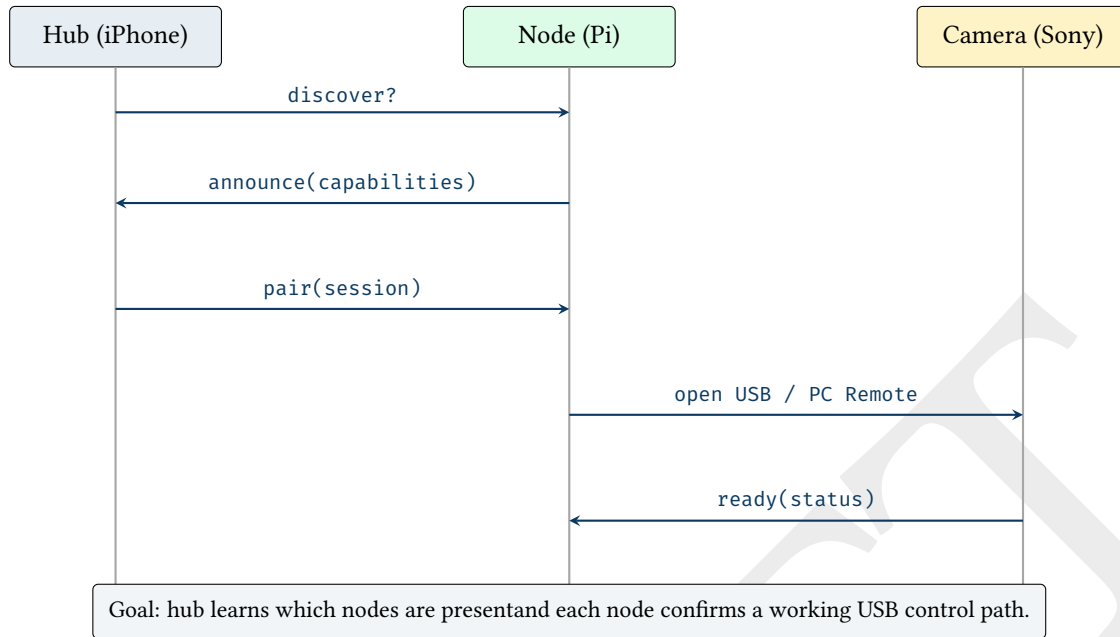


Figure 7. Sequence: bring-up and enrollment (conceptual protocol).

7.2 Scenario B: Focus Calibration at Dusk

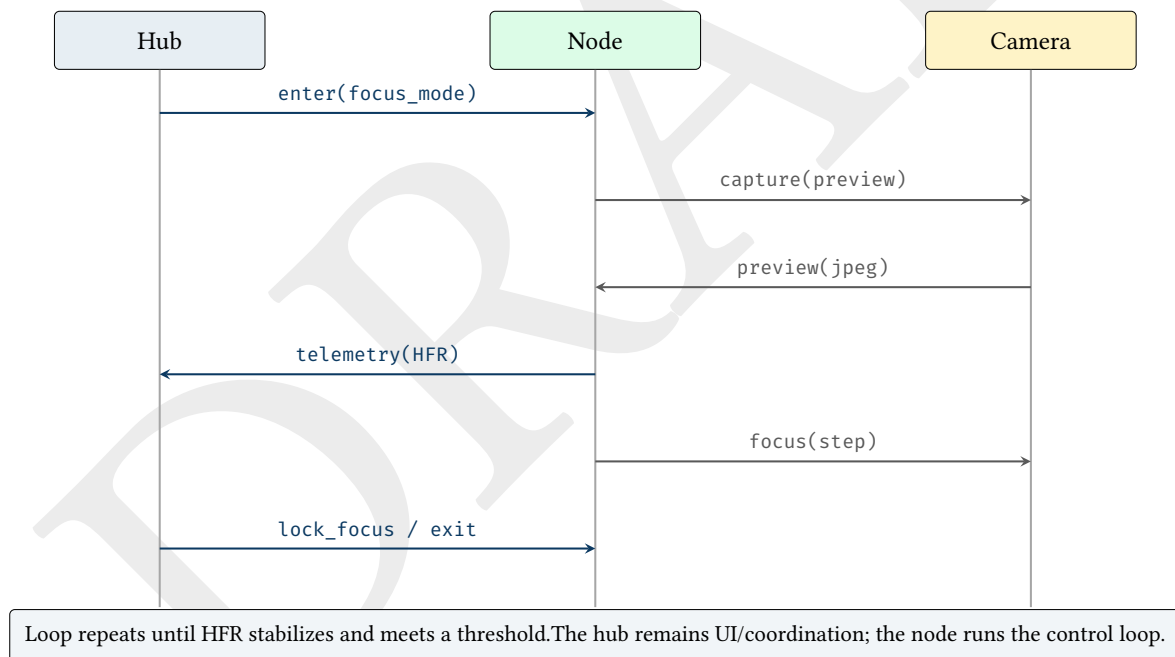


Figure 8. Sequence: focus calibration loop at dusk (conceptual).

7.3 Scenario C: Aurora Burst with Multi-node Synchronization

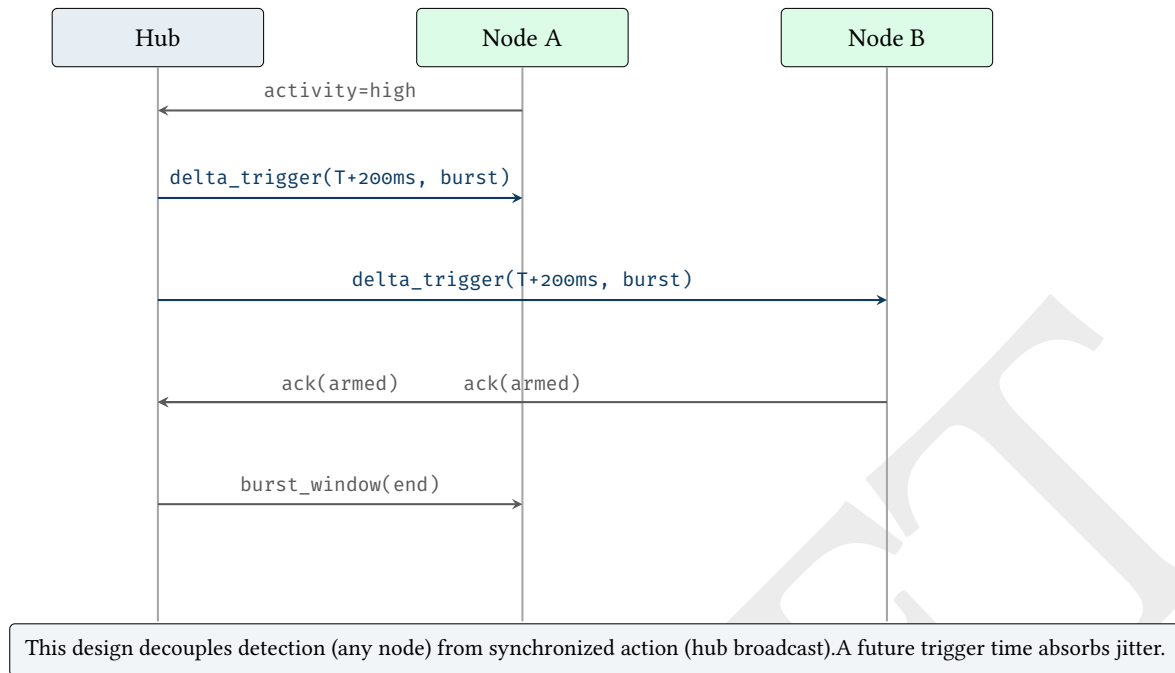


Figure 9. Sequence: one node detects activity; hub coordinates a synchronized burst.

8 Formal State Models

The system is safety-critical in the sense of opportunity cost: failure wastes an entire night of data. Clear state models simplify implementation and test harness design.

8.1 Capture Node State Machine

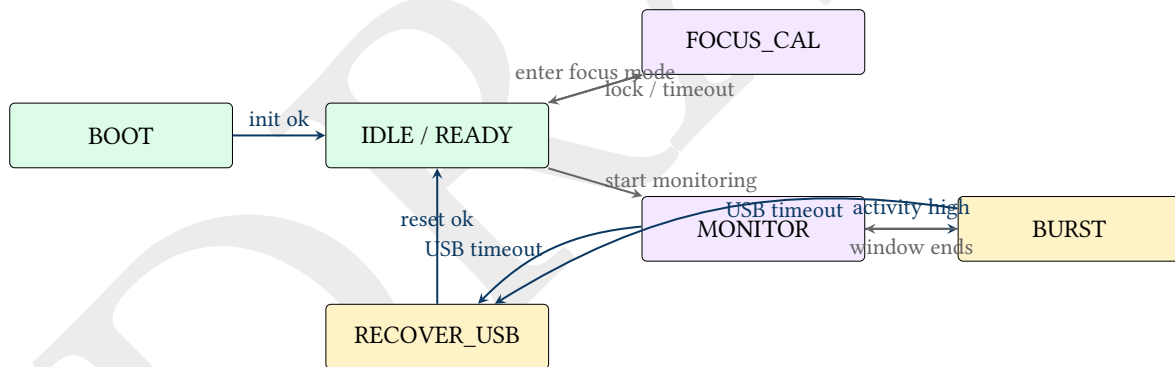


Figure 10. Conceptual node state machine including USB recovery as a first-class state.

8.2 Hub State Machine

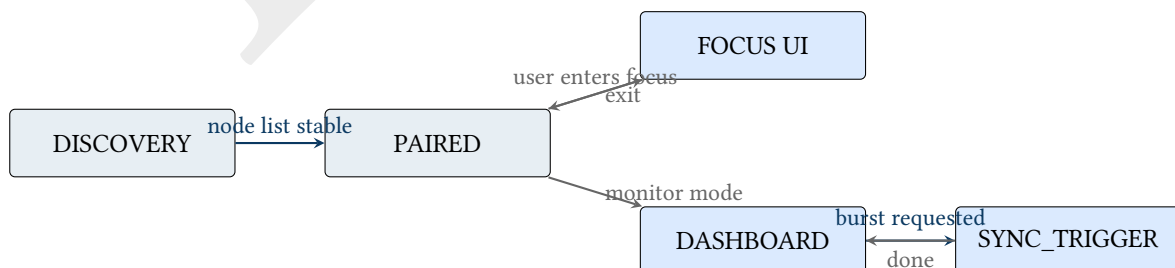


Figure 11. Conceptual hub state machine centered on coordination, not computation.

9 Timing and Performance Budgets

Although this report is conceptual, the control loops imply quantitative budgets. The key metric is closed-loop latency: the time from a preview frame being captured to a control action being applied.

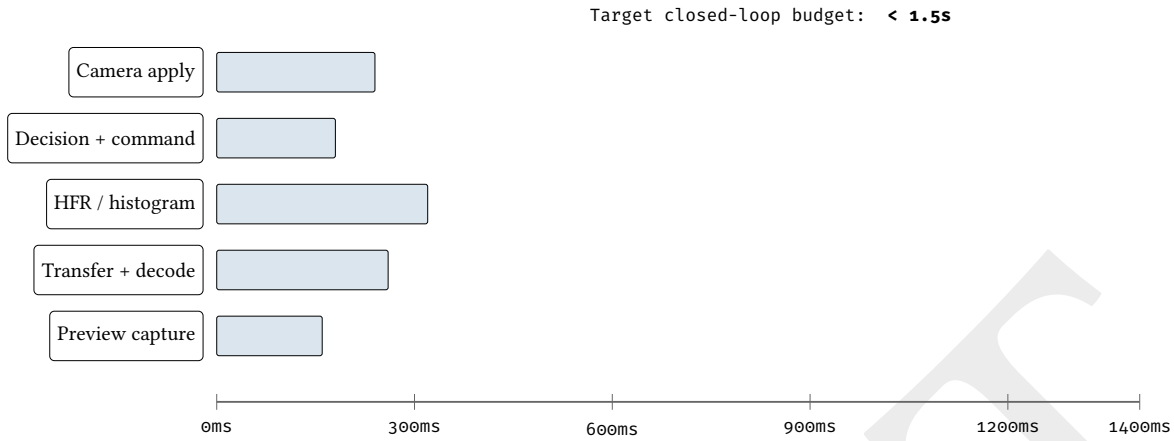


Figure 12. Conceptual latency budget for a single control iteration (focus/exposure).

9.1 Algorithmic Detail: HFR Estimation Pipeline (Per-frame)

HFR estimation is the core “advanced” perception primitive. The algorithm below is intentionally framed as a pipeline with uncertainty gates rather than a linear flowchart: each stage can emit confidence and failure reasons.

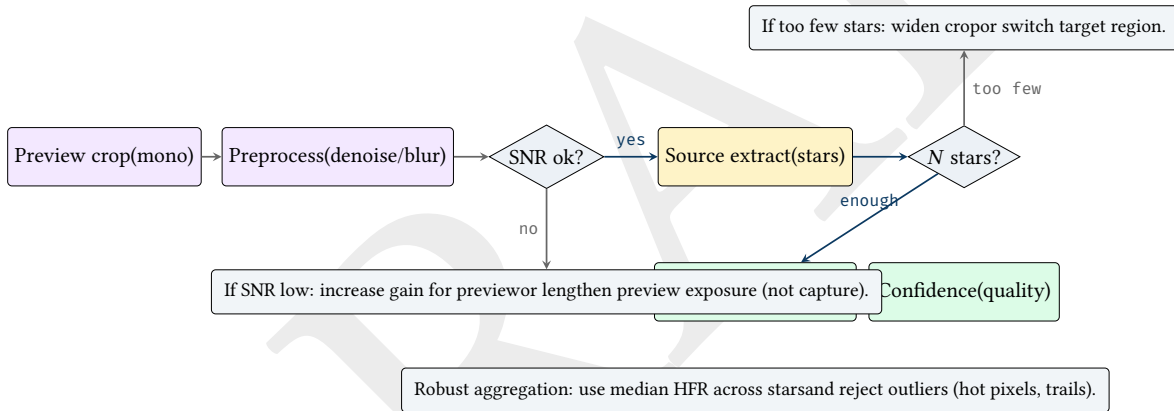


Figure 13. HFR estimation as a gated perception pipeline with explicit failure modes and confidence.

9.2 Algorithmic Detail: Burst Detection as a Change-point Problem

“Burst” events can be framed as change points in a time series of preview-derived signals (brightness, texture, motion, or ML scores). This motivates designs that use hysteresis, smoothing, and minimum-duration windows.

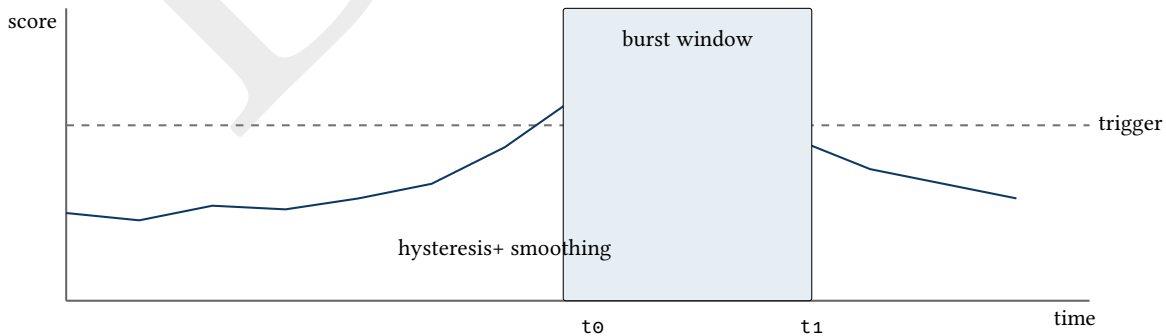


Figure 14. Burst detection as change-point detection on a scalar activity score with hysteresis and minimum-duration windows.

9.3 Algorithmic Detail: Focus Search Under Noise (Coarse-to-fine)

Autofocus for stars is an optimization under noisy measurements: each focus step yields a new preview frame and a stochastic HFR estimate. A robust strategy is a coarse-to-fine search: bracket the minimum with large steps, then refine with smaller steps and a stop condition based on improvement and measurement confidence.

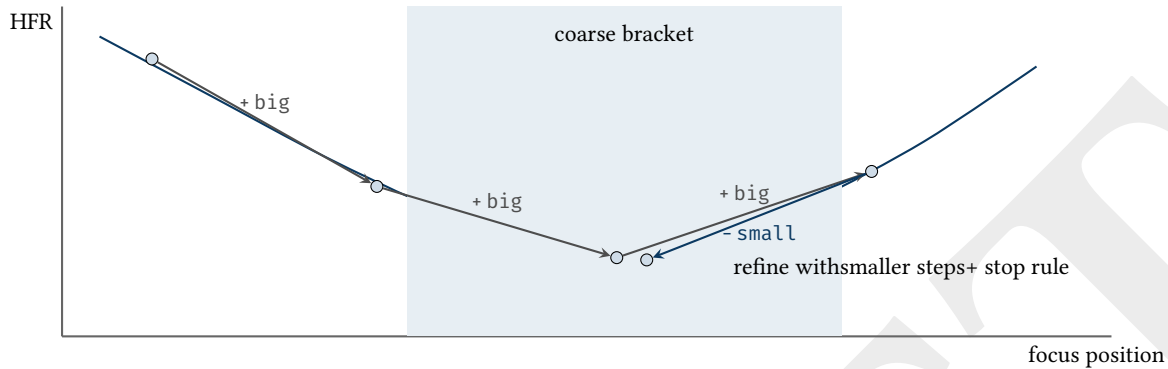


Figure 15. Focus search under noise: bracket the minimum with large steps, then refine locally with smaller steps and confidence-aware stopping.

9.4 Algorithmic Detail: Exposure Trade-offs as a Pareto Frontier

Aurora capture optimizes multiple competing objectives: avoid star trailing (short shutter), preserve beam structure (short shutter), avoid clipping (lower ISO/shutter), and avoid excessive noise (lower ISO / sufficient exposure). These are not simultaneously satisfiable; the system should operate on a Pareto frontier and switch regimes during bursts.

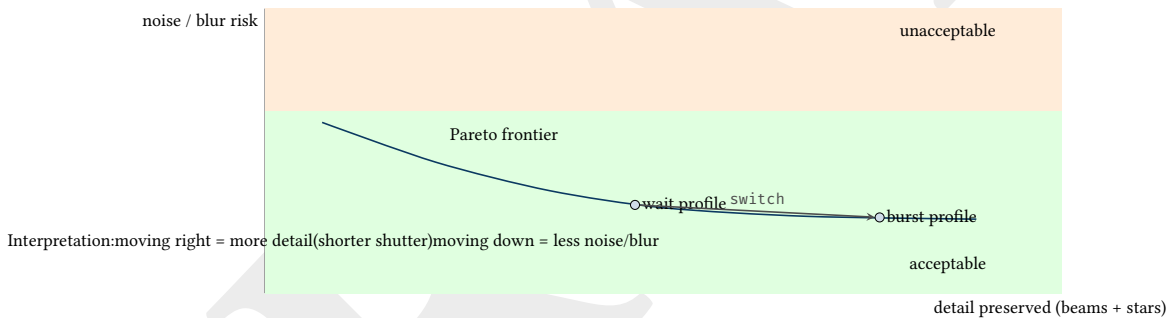
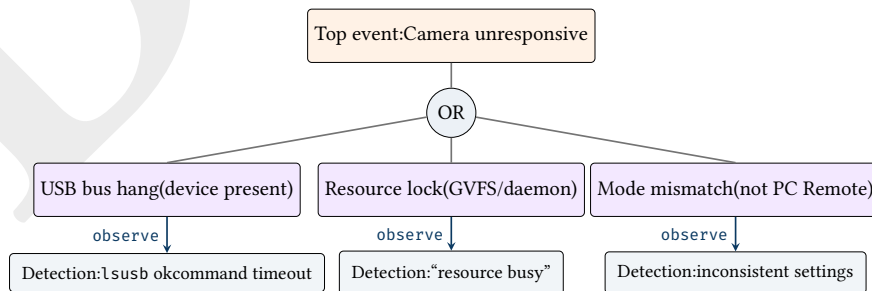


Figure 16. Multi-objective exposure tuning: operate along a Pareto frontier and switch operating points when activity changes.

9.5 Reliability Analysis: Fault Tree and Recovery Confidence

Many failures manifest identically (“no response”) but have different remedies. A lightweight fault tree helps separate detection signals from actionability and allows the system to emit a recovery confidence value.



Recovery actions:(1) power-cycle USB port(2) kill locking process(3) set PC Remote + reconnectEmit: **recovery confidence** based on which leaf fired.

Figure 17. Fault tree: separate failure causes that share the same symptom, and tie each cause to a specific recovery action and confidence.

10 Design Rules and Anti-patterns (Field-driven)

This section elevates domain pitfalls into explicit design constraints.

: No RAW in Feedback Loops

Do not rely on RAW transfer for any real-time decision. Preview-derived metrics are the only viable feedback signal in the field.

: Recovery is a State

USB hang recovery is not an exception path; it is an expected state transition with explicit detection, actuation (power-cycle), and a return-to-service criterion.

: Edge-heavy Perception

All perception that closes a loop (HFR, histogram, activity) executes on the node to avoid Wi-Fi latency and bandwidth coupling.

11 Implementation Roadmap (Conceptual)

Even without committing to a specific API stack, the work packages can be defined by which loop they enable and how they can be validated.

Phase	Primary Deliverable	Validation
P1	Node USB control + preview acquisition	Repeatable shutter/ISO/focus commands in cold soak
P2	HFR and histogram metrics	HFR correlates with visible star sharpness; no highlight clipping
P3	Hub UI + telemetry dashboard	Field setup in gloves; stable node status and simple triggers
P4	Burst profile switching + delta-triggering	Multi-node burst alignment within tolerance; minimal missed bursts

Table 3. Conceptual roadmap aligned to control loops and validation.

12 Assumptions, Invariants, and Testable Claims

This report is intentionally conceptual (no fixed API), but it still aims to be falsifiable. Each claim below implies a test harness and measurable acceptance criteria.

12.1 System Assumptions

1. A low-resolution preview stream is available at a cadence sufficient for control (order 1–5 Hz).
2. Focus steps are repeatable (monotonic mapping between command and focus motor motion).
3. Preview-derived metrics (HFR, histogram, activity score) correlate strongly with the desired outcomes (sharp stars, unclipped highlights, preserved beam structure).

12.2 Invariants

: Safety over Aggression

If the histogram indicates clipping risk, the system must de-rate exposure even if that reduces activity score sensitivity.

: Loop Independence via Priority

When control loops conflict, resolve by priority: (1) prevent clipping, (2) maintain focus, (3) optimize burst/detail.

12.3 Testable Claims

Claim	Statement	How to Test
C1	Closed-loop exposure adjustments occur within a bounded latency window under nominal conditions.	Inject step-change brightness and measure time-to-command.
C2	HFR estimates are stable enough to drive focus search without oscillation.	Run repeated focus sweeps and compute variance / convergence.
C3	Burst switching preserves beam detail compared to a fixed-profile baseline.	A/B capture sequences; compare clipped pixels and structure metrics.
C4	USB hang recovery returns the node to service without human intervention in most cases.	Cold soak + scripted fault injection; count successful recoveries.

Table 4. Representative claims and validation strategies.

13 Validation Protocols

This section proposes field-realistic protocols. The goal is not perfect lab repeatability; it is a disciplined way to learn which assumptions fail.

13.1 Protocol: Preview Cadence and Latency Budget

Measure the end-to-end time from preview capture to actuation. Record distributions (median, 95th percentile), not just means.

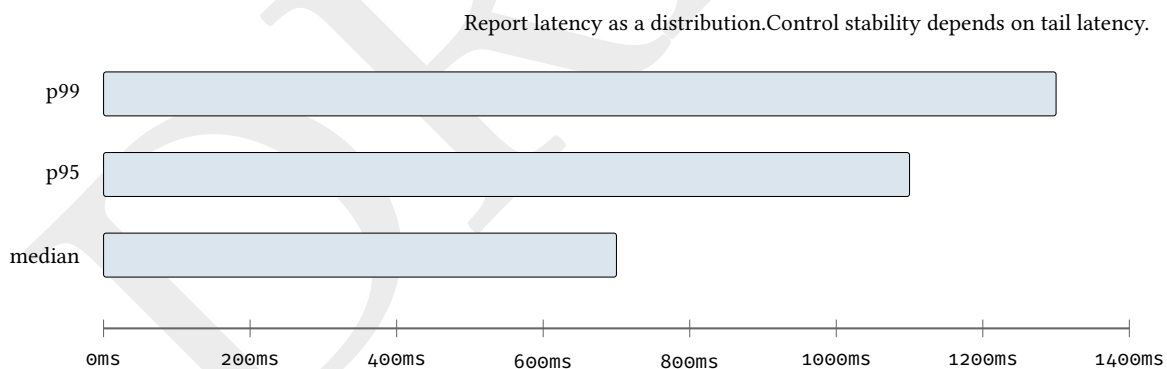


Figure 18. Latency reporting template (illustrative).

13.2 Protocol: Focus Drift and Refocus Triggers

Track HFR over time while temperature drops. Identify a drift threshold that predicts visible star degradation, then convert it into an alert or refocus trigger.

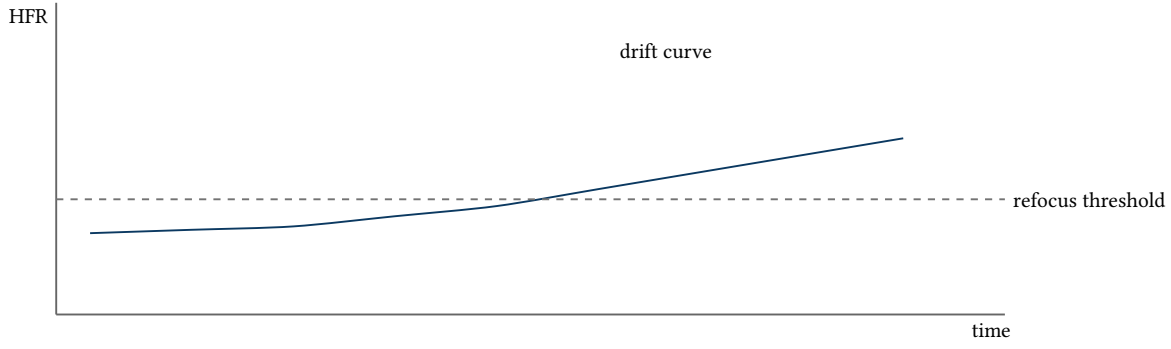


Figure 19. Focus drift protocol: estimate the threshold where refocus becomes necessary (illustrative).

13.3 Protocol: Burst Benefit A/B Test

Capture comparable sequences with (A) fixed exposure and (B) burst switching. Compare clipped pixels, median star sharpness, and a proxy for aurora structure preservation.

14 Open Questions and Research Risks

The following items are deliberately left as open research questions because their answers depend on real field performance.

1. Preview stream cadence/latency on target camera models under cold conditions.
2. The robustness of HFR estimates when aurora brightness overwhelms star contrast.
3. The best activity signal for burst switching in v1 (histogram heuristics vs texture/motion vs ML).
4. Multi-node synchronization tolerance required for downstream use (stitching vs simple multi-angle capture).

15 Estimation and Control Under Uncertainty

This section introduces a unifying mental model: the system maintains an internal belief about the scene and device health, then uses that belief to choose actions. The belief is not a full probabilistic model; it is a compact set of scores (HFR, histogram clip margin, activity score, device health) paired with confidence values.

15.1 Belief State and Confidence Gates

Design Decision 3: Confidence-first Control

The system should not act aggressively on low-confidence estimates. Confidence gates convert uncertain perception into conservative actions or requests for more evidence (more frames, wider crop, different preview settings).

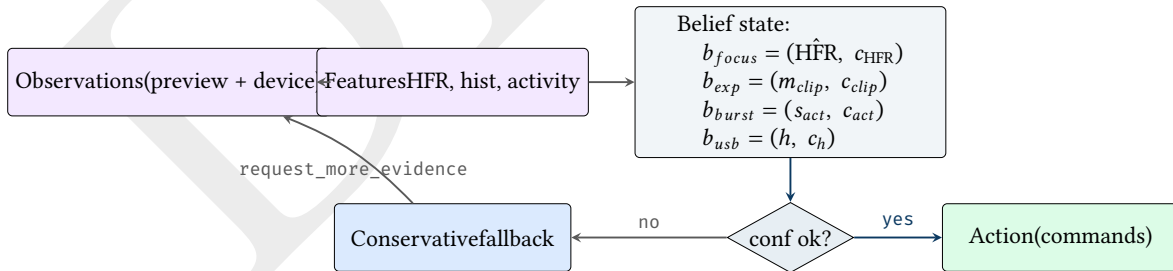


Figure 20. Belief-state framing: estimates are paired with confidence, and confidence gates determine whether to act or gather more evidence.

15.2 Uncertainty Propagation (Conceptual)

To avoid brittle behavior, confidence should decrease when conditions degrade (e.g., aurora overwhelms star contrast). The system can treat confidence as a function of SNR and star count.

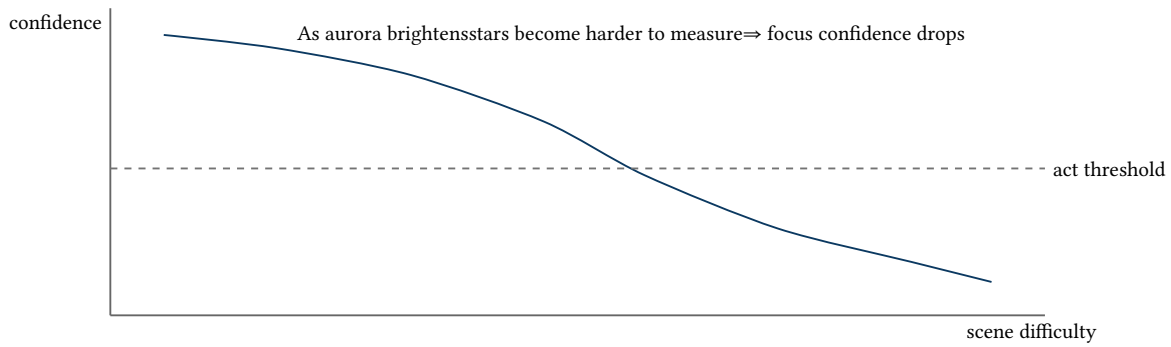


Figure 21. Confidence decreases with increasing scene difficulty; controllers de-rate actions below a threshold.

16 Test Harness Design

This section defines how to validate the system without relying on perfect aurora occurrences. The approach is to separate perception validation (offline datasets), control validation (synthetic perturbations), and reliability validation (fault injection).

16.1 Experiment Matrix (Fault Injection)

Fault	Detect	Recover	Assert	Score
USB hang	timeout	power-cycle	ready again	p_{rec}
Resource lock	busy	kill daemon	connect ok	p_{rec}
Low SNR	gate fail	needs tuning	safe mode	c

Key: treat each row as an executable test case. p_{rec} : recovery success probability estimate; c : confidence calibration.

Figure 22. Fault injection matrix template: each row defines a reproducible test with detection, recovery, and assertions.

16.2 Calibration: Recovery Confidence

Recovery confidence should be calibrated empirically. After repeated injections, the system can map detected leaf causes to observed recovery rates.

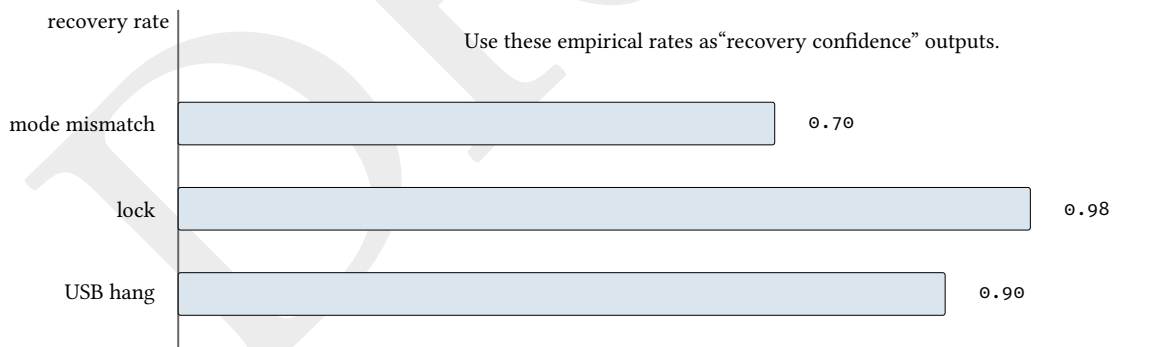


Figure 23. Recovery confidence calibration (illustrative): map detected fault classes to empirical recovery rates.

17 Multi-node Theory and Synchronization Error Budget

Multi-node capture is governed by the distribution of trigger misalignment. Misalignment is the sum of multiple independent (or partially correlated) sources.

17.1 Error Decomposition

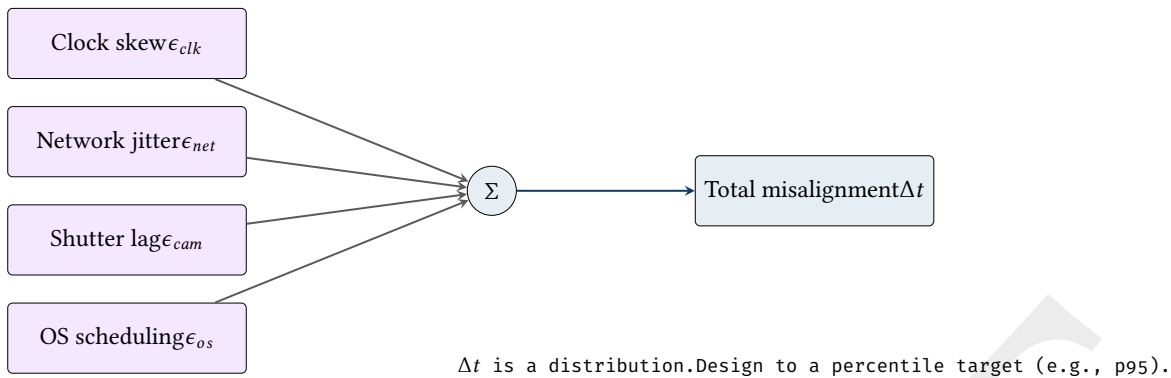


Figure 24. Synchronization error budget: decompose misalignment into clock, network, camera, and OS components.

17.2 Timing Diagram: Delta-trigger vs Immediate Trigger

Delta-triggering reduces sensitivity to network jitter by scheduling a future time.

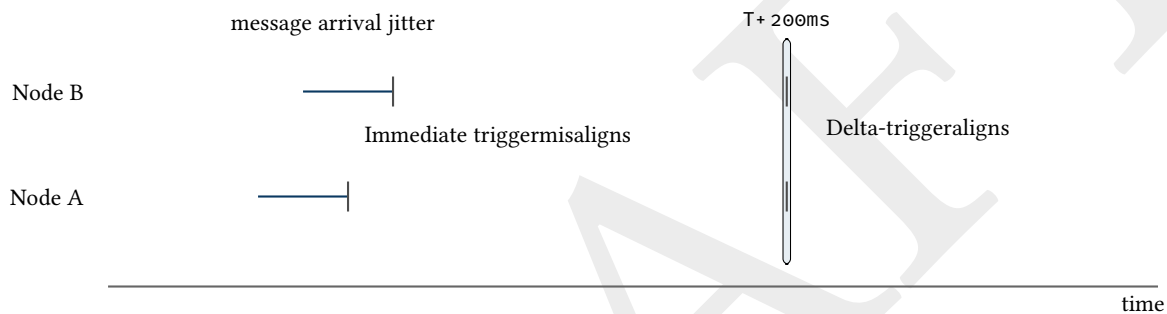


Figure 25. Delta-triggering aligns actions at a future timestamp despite message arrival jitter.

17.3 Tolerance Targets

Tolerance should be specified by downstream goals. For example, panorama stitching may tolerate looser sync than any attempt at parallax-based effects.

Use Case	Target (p95)	Notes
Multi-angle documentation	250 ms	Human perception tolerant; goal is capture coverage.
Panorama stitching (static scenes)	100 ms	Aurora motion may still dominate; depends on focal length.
Synchronized burst aesthetics	50 ms	More sensitive; requires tighter clocks and predictable shutter lag.

Table 5. Illustrative synchronization tolerance targets (to be validated).

The following requirements identifiers define v1 scope and support roadmap traceability.

Req ID	Description	Status
SHRP-01	Reliable USB camera control (ISO, shutter, aperture, trigger)	Pending
SHRP-02	Real-time HFR calculation for star sharpness monitoring	Pending
SHRP-03	Remote focus stepping via USB	Pending
SHRP-04	Programmatic recovery from Sony USB hangs	Pending
CAPT-01	Aurora burst logic (adaptive exposure profiles)	Pending
CAPT-02	Histogram-based exposure guard rails	Pending
CAPT-03	Intervalometer with interrupted sequence support	Pending
MOB-01	Low-latency iPhone viewfinder from node preview	Pending
MOB-03	Telemetry dashboard (battery/temp/HFR/health)	Pending
SYNC-01	Multi-node discovery and synchronized triggering	Pending

Table 6. Initial v1 requirements traceability table (imported from project planning).

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Revision History

Version	Date	Changes
0.1.0	March 2026	Initial technical report skeleton; architecture + control-loop diagrams

Todo list
